Definition (Symmetric stacking semicategory)
A *symmetric stacking semicategory* is an associative stacking semicategory $\mathbf{C}$ with:

<u>Constituents</u>

- Functions

$$\sigma_s : \mathrm{Hom}_{\mathbf{C}}(\otimes_{i=1}^{n} X_i; Y) \to \mathrm{Hom}_{\mathbf{C}}(\otimes_{i=1}^{n} X_{\sigma^{-1}(i)}; Y)$$

  and

$$\sigma_t : \mathrm{Hom}_{\mathbf{C}}(X; \otimes_{i=1}^{n} Y_i) \to \mathrm{Hom}_{\mathbf{C}}(X; \otimes_{i=1}^{n} Y_{\sigma(i)})$$

  for every $\sigma \in \mathbf{Perms}(n)$, $n \geq 1$. We call these functions *source permutations* and *target permutations*, respectively.

<u>Conditions</u>

- *Compatibility with permutation composition:*

  The equations

$$\sigma_s \fatsemi \pi_s = (\pi \fatsemi \sigma)_s \quad \text{and} \quad \sigma_t \fatsemi \pi_t = (\sigma \fatsemi \pi)_t$$

  hold for all $\sigma, \pi \in \mathbf{Perms}(n)$, $n \geq 2$.

- *Compatibility with permutation identities:*

  For any $n \in \mathbb{N}$, if $\sigma = 1_n$ is the identity permutation in $\mathbf{Perms}(n)$, then

$$(1_n)_s : \mathrm{Hom}_{\mathbf{C}}(\otimes_{i=1}^{n} X_i; Y) \to \mathrm{Hom}_{\mathbf{C}}(\otimes_{i=1}^{n} X_i; Y)$$

  and

$$(1_n)_t : \mathrm{Hom}_{\mathbf{C}}(X; \otimes_{i=1}^{n} Y_i) \to \mathrm{Hom}_{\mathbf{C}}(X; \otimes_{i=1}^{n} Y_i)$$

  are the identity functions.

- *Compatibility with block permutations:*

  For any $n \in \mathbb{N}$ and any partition $n = k_1 + \cdots + k_r$, if $Z_1, \ldots, Z_r$ are equal to the r factors in a "partitioned" stacking of the form

$$X = (X_1 \otimes \cdots \otimes X_{k_1}) \otimes (X_{k_1+1} \otimes \cdots \otimes X_{k_1+k_2}) \otimes \\ \cdots \otimes (X_{k_1+\cdots+k_{r-1}+1} \otimes \cdots \otimes X_n)$$

  then for any $\sigma \in \mathbf{Perms}(r)$, the source permutation map σ_s with respect to this r-fold stacking is equal to the source permutation map $(\sigma_*(k_1, \ldots, k_r))_s$ with respect to the n-fold stacking

$$X = X_1 \otimes \cdots \otimes X_n.$$

  (Here, $\sigma_*(k_1, \ldots, k_r)$ is the block permutation induced by σ and the partition $n = k_1 + \cdots + k_r$.)

  The analogous statement for target permutation maps is also required to hold.

- *Compatibility with semicategory composition:*

For all $f \in \mathrm{Hom}_{\mathbf{C}}(X; \otimes_{i=1}^{n} Y_i)$ and $g \in \mathrm{Hom}_{\mathbf{C}}(\otimes_{i=1}^{n} Y_{\sigma(i)}; Z)$

$$\sigma_t(f) \fatsemi g = f \fatsemi \sigma_s(g).$$

▹ *Compatibility with stacking:*

For all $f \in \mathrm{Hom}_{\mathbf{C}}(\otimes_{i=1}^{n} X_i; Y_1)$, $g \in \mathrm{Hom}_{\mathbf{C}}\left(\otimes_{i=n+1}^{n+m} X_i; Y_2\right)$,

$$\sigma_s(f) \otimes \pi_s(g) = (\sigma \otimes \pi)_s(f \otimes g)$$

holds for all $\sigma \in \mathbf{Perms}(n), \pi \in \mathbf{Perms}(m)$, where $\otimes$ is the stacking operation in **Pm**.

Similarly, for all $f \in \mathrm{Hom}_{\mathbf{C}}(X_1; \otimes_{i=1}^{n} Y_i)$ and $g \in \mathrm{Hom}_{\mathbf{C}}\left(X_2; \otimes_{i=n+1}^{n+m} Y_i\right)$,

$$\sigma_t(f) \otimes \pi_t(g) = (\sigma \otimes \pi)_s(f \otimes g)$$

holds for all $\sigma \in \mathbf{Perms}(n), \pi \in \mathbf{Perms}(m)$.

▹ *Sliding:*

Given $f_i : X_i \to Y_i$, $1 \leq i \leq n$, it holds that

$$\sigma_t(\otimes_{i=1}^{n} f_i) = \sigma_s(\otimes_{i=1}^{n} f_{\sigma(i)})$$

and

$$\sigma_s(\otimes_{i=1}^{n} f_i) = \sigma_t(\otimes_{i=1}^{n} f_{\sigma^{-1}(i)}).$$